

ADAMAS: Cryptographically Binding a Diamond’s Physical Signature to Its On-Chain Provenance

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Abstract

Distributed ledgers give diamond provenance systems tamper-evident records. They do not give those records any verifiable connection to the physical stones the records describe. Every deployed platform closes that gap with a trusted operator: a laboratory scans a stone, an algorithm declares a match against a stored template, and the ledger faithfully records the operator’s assertion. An adversary who substitutes stones between checkpoints, compromises a scanner, or replays a stored measurement defeats the system while leaving the chain internally consistent. We call this the *binding gap*.

We present ADAMAS, a protocol that derives a cryptographic key directly from a stone’s crystallographic defect structure and makes possession of the physical stone a prerequisite for every provenance operation. ADAMAS has three layers: a learned $SE(3)$ - and permutation-invariant embedding of the internal inclusion constellation that provides alignment-free screening; a registration-and-vault stage that converts the inclusion point set into a reproducible key; and a challenged-interrogation layer over photoluminescence and nitrogen-vacancy sub-volumes that defeats replay and constrains a compromised sensor. Custody transfers are co-signed by the stone itself, so a record cannot move without the object.

We derive a feasibility bound relating bit error rate, per-bit min-entropy, and extractable key length, and evaluate the pipeline on a synthetic population of 1,000 simulated stones. Three findings shape the design. First, learned within-class whitening reduces the screening equal error rate from 8.9×10^{-2} to 1.9×10^{-2} , but the resulting alignment-free descriptor carries only ≈ 55 bits of extractable entropy, far below what a 128-bit binding requires; a single-stage fuzzy extractor over global invariants is structurally insufficient. Second, geometry alone cannot separate natural from laboratory-grown stones (200/200 synthetic CVD substitutes pass the geometric screen), whereas the spectroscopic block separates them at 99.8% accuracy: the material-class gate must be spectroscopic. Third, the registered point-set vault reaches 90-bit security at a 97% unlock rate with 14 mapped inclusions, and 128-bit security requires reliably mapping ≥ 20 inclusions per stone, which is an instrument specification rather than a protocol parameter. We also find that an adversary holding 90% of a stone’s inclusion map unlocks the vault with probability 0.49, which makes the routine publication of clarity plots a live disclosure risk.

We argue that unforgeability in the absolute sense is

unattainable, and that the achievable goal is a reduction: forgery becomes equivalent to atomic-scale replication of a defect constellation, compromise of an attested instrument, or possession of the genuine stone, with each made detectable and attributable. All evaluation is simulation on synthetic data; we specify the physical study with graded stones that would validate or refute it.

1 Introduction

In 2024 a gemological laboratory received a six-carat stone bearing the girdle inscription of a certified natural diamond. The inscription was authentic in the sense that it matched a real report. The stone was not: photoluminescence (PL) spectroscopy revealed a silicon-vacancy doublet near 737 nm, diagnostic of chemical vapor deposition growth. The substitution was caught because a laboratory happened to run spectroscopy, and because the observed inclusions disagreed with the plot on the referenced report. Absent that scrutiny, a digitally impeccable provenance record would have accompanied the wrong object [28].

That incident is the whole problem in miniature. A provenance record is a claim about a physical object. Cryptography and consensus can make the claim tamper-evident, ordered, and attributable. Neither can make the claim *true*. The step where a measurement of matter becomes a statement in a ledger is the step where an adversary operates, and it is the step that distributed-ledger provenance systems have historically delegated to a trusted operator.

Why now. The economic pressure on this seam is increasing sharply. Laboratory-grown stones are visually and gemologically close to natural ones, sell for a fraction of the price, and have taken a large share of the engagement market, while natural rough prices have fallen for several consecutive years [32]. The premium attached to a natural-origin certificate is therefore both larger in relative terms and easier to counterfeit, since the counterfeit substrate is cheap and abundant. In parallel, the industry has consolidated around ledger-based provenance: De Beers launched Tracr in 2018, opened it to the wider industry in 2023, and in May 2026 the Gemological Institute of America (GIA) agreed to acquire a 30% shareholding as the platform moves toward independence [29, 30]. More than five million rough diamonds have been registered at source, and since January 2025 all newly sourced De Beers rough of one carat and above is regis-

tered [29]. The infrastructure that a binding attack would target is now real, large, and load-bearing.

The binding gap. Deployed systems bind records to stones by *algorithmic matching under a trusted operator*. Sarine’s Diamond Journey, for example, scans a rough stone, plans the cut, re-scans the polished result, and confirms identity by matching proportions, inclusions, and other features against the recorded plan; the registration then transfers to the new owner [35, 34]. This is a genuine technical achievement, and it is exactly what the industry needs for logistics. As a security mechanism it has four structural properties an adversary can use:

1. **The comparison is a similarity threshold, not a key.** A match produces a boolean from a trusted process. Nothing cryptographic is derived from the stone, so nothing the stone produces can be independently verified by a third party.
2. **The template must be stored in the clear** to be compared against, so whoever holds the registry holds every fingerprint, and a registry breach is a permanent, unrevocable compromise.
3. **There is no freshness.** A stored scan replayed by a compromised or merely lazy scanner is indistinguishable from a live measurement.
4. **The stone never signs anything.** Custody transfers are authorized by parties, so a record can move while the object stays still, or vice versa.

What is actually achievable. We state our position plainly, because the framing of the problem invites overclaiming. A protocol cannot make physical provenance unforgeable. Any system that turns matter into bits contains a sensor, and a sufficiently compromised sensor can assert anything. What a protocol *can* do is reduce forgery to a small set of irreducible capabilities and make each one expensive, detectable, and attributable. ADAMAS targets the reduction in Figure 1: to forge provenance an adversary must either replicate a defect constellation at atomic scale, compromise an attested instrument and defeat its challenge-response, or physically hold the genuine stone. The third case is uninteresting, since an adversary holding the genuine stone would rather sell it.

Contributions.

1. **A threat model for physical-digital binding** in diamond provenance (Section 3), covering substitution, cloning, replay, oracle compromise, template inversion, and re-cut laundering, with explicit security goals.
2. **A three-layer signature architecture** (Section 4) that assigns each physical modality the role its entropy and cost actually support: inclusion geometry for identity, spectroscopy for material class, nitrogen-vacancy sub-volumes for the root of trust and for challenge-response.
3. **A feasibility bound** (Proposition 1) relating bit error rate p , per-bit min-entropy rate ρ , and extractable key length, which converts “use machine learning and cryptography”

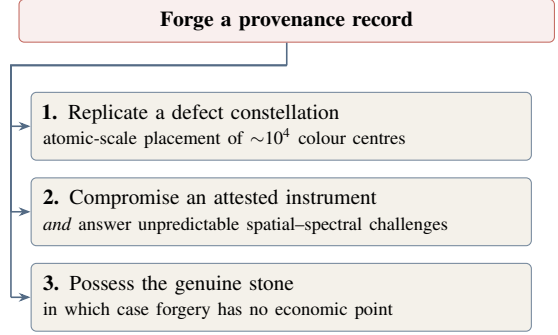


Figure 1: The reduction ADAMAS targets. The goal is to eliminate every forgery path except three irreducible ones, and to make the first two detectable.

into a hard joint requirement on the embedding and shows where the design cliff sits.

4. **A binding protocol** (Section 7) in which the stone holds a keypair, every custody transfer is co-signed by the stone, verification is a nonce-bound challenge-response over an adversary-unpredictable sub-volume, and re-polishing is handled by an explicit key-transition ceremony rather than by pretending the fingerprint is invariant.
5. **An evaluation on 1,000 simulated stones** (Section 9) producing three findings that constrain any system of this type: the alignment-free descriptor is structurally insufficient for cryptographic binding; the natural-versus-grown gate must be spectroscopic rather than geometric; and 128-bit security requires mapping ≥ 20 inclusions per stone, an instrument requirement we state as a specification for the physical study.

On the status of the evaluation. We have no access to physical stones. Every number in Section 9 comes from a generative model whose assumptions we state in full and whose code we release. The results are therefore predictions about a measurement campaign, not measurements. We regard the generative model as the principal threat to validity and devote Section 11 to it. Section 12 specifies the study with graded stones that would settle the question.

2 Background and Related Work

2.1 Diamond defects as an entropy source

A natural diamond accumulates structure over geological time that no manufacturing process reproduces. Four families of that structure are gemologically accessible.

Internal inclusion constellations. Crystals, feathers, clouds, and pinpoints occupy stochastic three-dimensional positions determined by growth history. Commercial mapping systems already recover these: Sarine’s Galaxy line produces 3D inclusion maps of rough stones at stated micron-level resolution [33, 34], and GIA clarity plots record the same information in graded form. This modality is the one the

industry can measure today at scale, which is why we treat it as the identity layer.

Photoluminescence spectra. Under laser excitation at liquid-nitrogen temperature, optically active defects emit at characteristic zero-phonon lines: NV^0 at 575 nm, NV^- at 637 nm, GR1 at 741 nm, 3H at 503.5 nm, and the silicon-vacancy $[SiV]^-$ doublet at 736.6/736.9 nm [23, 19]. PL is far more sensitive than absorption and is central to laboratory determination of growth origin and treatment history [23, 24]. The $[SiV]^-$ doublet is frequently incorporated during CVD growth through etching of silicon-bearing chamber components and is rare in natural stones [27, 21]; the 596/597 nm doublet has been reported in as-grown CVD material and not in natural diamond [25, 22]. Deep-ultraviolet fluorescence imaging resolves growth striations characteristic of CVD [26].

Two cautions apply. Diagnostic features are being suppressed as growth techniques improve, and GIA has documented CVD material in which the usual markers are minimized or absent [26]. Post-growth HPHT annealing can also create optically active $[SiV]^-$ in natural stones that previously showed none [24]. Growth-origin determination by PL is consequently an arms race, and any architecture that stakes identity on it inherits that fragility. We stake only material *class* on it.

Nitrogen-vacancy ensembles. At the single-defect level, the positions, charge states, and crystallographic orientations of individual NV centres constitute a vastly larger reservoir. NV physics is well characterized because of its use in quantum sensing and memory [16, 17, 18]. Cloning at this level would require deterministic atomic-scale placement of $\sim 10^4$ defects in a specified arrangement, which is beyond any demonstrated capability. Readout is slow and instrument-heavy, so we assign this modality to the root of trust and to high-value challenges rather than to routine verification.

Strain and type classification. Birefringence patterns and nitrogen aggregation state (types Ia, Ib, IIa, IIb) further constrain origin [20]. We use these as auxiliary class evidence.

2.2 Physical unclonable functions and fuzzy cryptography

The idea of deriving identity from irreproducible physical disorder originates with optical physical one-way functions [1] and silicon PUFs [2, 3]. Converting a noisy physical measurement into a stable cryptographic key is the province of fuzzy extractors [4], fuzzy commitment [5], and, for unordered point sets, the fuzzy vault [6]. Soft-decision helper data improves the reliability-leakage tradeoff substantially [7]. Modelling attacks are the standing counterweight: PUFs whose challenge-response behaviour admits a learnable model can be cloned in software [8]. Fuzzy vaults in particular are vulnerable to correlation attacks that distinguish chaff from

Property	Insc.	Tracr	Match	Label	ADAMAS
Tamper-evident record	—	✓	✓	—	✓
Intrinsic to the stone	—	—	✓	—	✓
Resists removal	—	n/a	✓	—	✓
Yields a key	—	—	—	✓	✓
Template kept secret	n/a	—	—	—	✓
Replay-resistant	—	—	—	—	✓
Stone co-signs transfer	—	—	—	—	✓
Clone detection	—	—	—	—	✓
Grown-material gate	—	—	partial	n/a	✓

Table 1: Capability comparison. *Insc.*: laser girdle inscription. *Tracr*: ledger provenance from source. *Match*: deployed inclusion-matching pipelines under a trusted operator. *Label*: affixed PUF tags including fabricated-nanodiamond labels.

genuine points across multiple enrolments [9], which constrains how we may re-enroll.

Diamond-based PUFs exist, but they occupy a different problem. Recent work uses *fabricated* diamond microparticles with silicon-vacancy centres, or fluorescent nanodiamond arrays authenticated by deep metric learning, as anti-counterfeiting labels affixed to goods [36, 37]. Those are labels: an adversary who removes the label defeats the system. Our setting inverts this. The gem itself is the PUF, so there is nothing to remove, and the adversary’s task becomes substituting the object rather than the tag.

2.3 Ledger provenance and the oracle problem

Consensus protocols guarantee integrity of recorded data [11, 12]. Bridging to off-chain facts is the oracle problem, and authenticated data feeds address it for digital sources by attesting to the provenance of the data rather than its truth [13]. Physical facts are harder: no attestation of a sensor’s software integrity establishes that the sensor was pointed at the claimed object. Privacy-preserving verification of on-chain predicates is well developed [14, 15], and we use it for the disclosure-minimizing parts of the protocol.

2.4 Positioning

Table 1 places ADAMAS against the alternatives. The distinguishing properties are that the stone holds a keypair, that no template is stored in the clear, and that verification is fresh.

3 Threat Model and Security Goals

3.1 Adversary classes

$\mathcal{A}_{\text{mkt}}$ **Market adversary.**

Holds stones (natural, grown, treated), can buy graded stones, can re-cut, has access to commercial scanning equipment and to any publicly published report data including clarity plots. Cannot compromise a certified instrument.

$\mathcal{A}_{\text{ins}}$ **Insider.**

Additionally controls one measurement endpoint: a scan-

ner’s firmware, an operator, or a laboratory workstation. Can submit chosen measurements.

$\mathcal{A}_{\text{reg}}$ **Registry adversary.**

Additionally reads all stored helper data and all on-chain state. Models a registry breach.

$\mathcal{A}_{\text{fab}}$ **Fabrication adversary.**

Additionally commands state-of-the-art CVD/HPHT synthesis and ion implantation, and may attempt to engineer a stone toward a target signature.

We assume the ledger’s consensus is honest; ledger-level attacks are orthogonal and addressed by standard means. We assume at least one honest certifying laboratory exists, and that hardware roots of trust in attested scanners are not extracted.

3.2 Attacks

A_1 **Record forgery.**

Create a provenance record for a stone never certified, e.g. to launder sanctioned or conflict-origin material.

A_2 **Substitution.**

Bind an existing valid record to a different physical stone. This is the 2024 incident [28] and the dominant economic attack.

A_3 **Cloning / double registration.**

Present two physical stones under one record, or register one stone twice.

A_4 **Replay.**

Re-use a previously captured measurement to authenticate without the stone present.

A_5 **Oracle compromise.**

A compromised instrument reports adversary-chosen measurements.

A_6 **Template inversion.**

Recover the fingerprint from public helper data, enabling A_2 or A_4 offline.

A_7 **Re-cut laundering.**

Re-polish a stone so its recorded identity no longer applies, then re-enroll it under a fabricated history; or re-polish a grown stone toward a natural stone’s recorded geometry.

A_8 **Enrolment poisoning.**

Corrupt the enrolment measurement so the recorded key belongs to the wrong object from the start.

3.3 Goals

G_1 **Binding.**

A verifier accepts a record for stone S only if S was physically measured during the verification.

G_2 **Freshness.**

Acceptance requires a measurement bound to a verifier-chosen nonce and challenge.

G_3 **Possession-gated transfer.**

A custody transfer is valid only with a live measurement of the stone.

G_4 **Confidentiality.**

Public data reveals negligible information about the fin-

gerprint; Section 9 shows this goal has real teeth.

G_5 **Clone detectability.**

Two entities presenting one identity produce an observable, attributable conflict.

G_6 **Lifecycle continuity.**

Legitimate re-polishing is representable as an authenticated key transition rather than a break.

4 The Physical Signature Layer

4.1 Modality tiering

A single modality cannot serve every role, because entropy, cost, and robustness trade against one another. ADAMAS assigns three roles.

M_1 **Inclusion geometry (identity).**

The 3D constellation of internal features. Measurable today at scale, robust to handling, and partially robust to re-polishing. Carries $O(10^2)$ extractable bits.

M_2 **Spectroscopy (material class).**

PL line intensities and ratios, IR type classification, deep-UV fluorescence morphology. Carries little *identity* entropy, because stones from one geological or growth population share spectra, but decides natural versus grown versus treated with high accuracy.

M_3 **NV sub-volume (root of trust, challenge).**

Confocal maps of individual colour centres within a designated sub-volume. Carries $> 10^4$ bits, is the only layer with credible resistance to $\mathcal{A}_{\text{fab}}$, and supports an enormous challenge space. Slow and instrument-heavy, so reserved for certification and high-value checkpoints.

4.2 Entropy budget

Consider a stone of characteristic half-dimension $R = 2.5$ mm containing K mappable inclusions localized to an effective cell size δ . The number of distinguishable cells is $N_c = (2R/\delta)^3$, and the unordered set of K positions carries at most

$$\mathbf{H}_\infty(M_1) \leq K \log_2 N_c - \log_2 K! \quad (1)$$

bits. The relevant δ is set by *registration* error rather than by the mapper’s raw localization precision, since the coordinate frame must be recovered from the same noisy point set. Table 2 evaluates Equation (1) and contrasts it with what we actually extract.

The table states the central engineering fact: the mapper’s precision is not the binding constraint. Frame recovery is. An adversary and a verifier both see the inclusion set only up to an unknown rigid transform, and every bit of angular uncertainty in that transform costs position precision proportional to radius.

Modality	Ideal $\mathbf{H}_\infty$	Extracted	Ratio
M_1 , raw $\delta=8\mu\text{m}$	~ 1100	—	—
M_1 , registered $\delta=91\mu\text{m}$	260	90	0.35
M_1 , alignment-free	—	55	—
M_2 , PL line ratios	~ 40	≈ 1	—
M_3 , NV sub-volume	$> 10^4$	n/s	—

Table 2: Entropy budget, in bits, for $K = 18$. The gap between raw localization and registration-limited precision is the single largest loss in the system and is the reason Section 6 spends its effort on registration. “n/s” marks a modality we do not simulate.

5 Stage A: Learned Invariant Embedding

5.1 Design requirements

Stage A maps a raw scan to a short binary code without reference to any stored template, so it can index a registry of millions and reject gross mismatches before the expensive stage runs. It must be invariant to instrument pose, invariant to the ordering of detected inclusions, tolerant of missed and spurious detections, and discriminative across stones.

5.2 Invariance by construction

Rather than teaching invariance through augmentation, we build it into the representation. Let a scan be $\{(x_a, w_a)\}_{a=1}^K$ with positions $x_a \in \mathbb{R}^3$ and size weights w_a normalized to sum to one. We use two complementary families.

Frame-free invariants. Functions of the pairwise distance matrix $D_{ab} = \|x_a - x_b\|$ are invariant to $SE(3)$ and, when symmetrized, to permutation. We use a size-weighted kernel density estimate of the normalized pairwise distances, a radial profile about the weighted centroid, the normalized inertia spectrum, and the leading eigenvalues of the affinity matrix $G_{ab} = w_a w_b \exp(-D_{ab}^2/2h^2)$ at two bandwidths. Distances are normalized by their median so that mild scale changes do not disturb the descriptor.

Canonical-frame occupancy. We diagonalize the size-weighted inertia tensor, resolve axis signs by the weighted third moment, and evaluate a Gaussian-splatted occupancy field on a 6^3 grid in the resulting frame. This block is far more informative than the frame-free family when the frame is recovered correctly, and degrades sharply when it is not, which is precisely the behaviour Section 6 must fix.

Concatenating these with absolute-scale statistics and the PL block gives a 326-dimensional raw descriptor.

5.3 Learned projection

The raw descriptor has strongly anisotropic noise and several exact linear dependencies, since normalized histograms lie on

a simplex. We therefore (i) reduce to the top d' principal directions of the total covariance, discarding the rank-deficient subspace; (ii) apply within-class covariance normalization, estimated from repeated scans of training stones, which is the supervised step that makes bits reliable; and (iii) order the resulting directions by between-class variance. In a deployed system this linear pipeline is the degenerate case of a trained encoder; we use the linear form so that the evaluation isolates the effect of within-class whitening rather than of network capacity. Section 12 specifies the $E(3)$ -equivariant graph encoder that should replace it once real data exists [38, 39, 40].

5.4 Reliability-ranked quantization

For direction j let $\sigma_{b,j}$ and $\sigma_{w,j}$ be between-stone and within-stone standard deviations, and $\text{SNR}_j = \sigma_{b,j}/\sigma_{w,j}$. For a median-thresholded bit, the flip probability of a stone drawn from the population is

$$p_j = \frac{1}{\pi} \arccos\left(\frac{\text{SNR}_j}{\sqrt{1 + \text{SNR}_j^2}}\right), \quad (2)$$

so $\text{SNR} = 1$ gives $p = 0.25$ and $\text{SNR} = 4$ gives $p = 0.078$. We allocate $k_j \in \{0, 1, 2, 3\}$ bits per direction by thresholding SNR_j at 1, 2, 4, quantize at population quantiles so each bit is balanced by construction, and Gray-code the levels so that a one-level error costs one bit. Bits are then ranked by empirical flip rate plus a bias penalty, and the top n are retained.

5.5 A feasibility bound

The following makes explicit what the embedding must deliver for the cryptographic stage to be possible at all. Let $W \in \{0, 1\}^n$ denote the selected bit block with per-bit min-entropy rate $\rho = \mathbf{H}_\infty(W)/n$, let p be the bit error rate between enrolment and probe, and let $h(\cdot)$ be the binary entropy function.

Proposition 1 (Feasibility). *A secure sketch built from a linear $[n, k]$ code that corrects a fraction p of errors, followed by a strong extractor with security parameter ϵ , yields a key of length at most*

$$\ell \leq n\rho - (n - k) - 2\log_2(1/\epsilon). \quad (3)$$

Reliable decoding requires $n - k \geq \gamma n h(p)$ for a finite-length margin $\gamma > 1$. Hence a key of length ℓ requires

$$n \geq \frac{\ell + 2\log_2(1/\epsilon)}{\rho - \gamma h(p)}, \quad \rho > \gamma h(p). \quad (4)$$

Sketch. The syndrome of a linear code leaks at most $n - k$ bits, so $\tilde{\mathbf{H}}_\infty(W \mid \text{SS}(W)) \geq \mathbf{H}_\infty(W) - (n - k)$ [4]. A strong extractor converts average min-entropy m into $m - 2\log_2(1/\epsilon)$ bits that are ϵ -close to uniform. Substituting the Shannon-limit rate with a finite-length penalty γ and rearranging gives Equation (4). $\square$

BER p	$h(p)$	required n for $\ell = 128$		
		$\rho=0.5$	$\rho=0.7$	$\rho=0.9$
0.05	0.286	1638	719	461
0.10	0.469	—	1866	760
0.15	0.610	—	—	1523
0.20	0.722	—	—	7600

Table 3: Proposition 1 evaluated at $\varepsilon = 2^{-64}$, $\gamma = 1.2$. Dashes mark infeasibility at any block length.

Remark 1. The condition $\rho > \gamma h(p)$ is a cliff, not a gradient. At $\ell = 128$, $\varepsilon = 2^{-64}$, $\gamma = 1.2$: with $\rho = 0.7$, a bit error rate of 5% needs $n \geq 719$ while 10% needs $n \geq 1866$ and 13% is infeasible at any n . Driving down BER and driving up per-bit entropy are therefore not interchangeable objectives, and a loss that trades one for the other is mis-specified. This is why the embedding must be trained with an explicit bit-decorrelation term alongside the reliability term.

Table 3 tabulates the requirement, and Figure 5 plots the feasible region with our measured operating point.

6 Stage B: Registration and Key Extraction

Section 9 will show that Stage A cannot supply a 128-bit key. Stage B therefore abandons alignment-free operation and instead recovers the frame explicitly, which is what unlocks the entropy in Table 2.

6.1 Registration

Canonical-frame estimation from a sparse, partially observed point set is unreliable in exactly the way that matters: when inertia eigenvalues are close, the axis *ordering* swaps between scans, and sign disambiguation by third moments fails when a single spurious detection sits far from the centroid. A naive canonical frame therefore succeeds most of the time and fails catastrophically the rest of the time, producing a bimodal recovery distribution that no threshold handles well.

We resolve this by hypothesis search. Both point sets are placed in canonical coordinates; the probe is then transformed by each of the 24 proper signed permutation matrices, and each hypothesis is refined by iterative closest point with an annealed correspondence gate ($600 \rightarrow 120 \mu\text{m}$) using a similarity Procrustes step that permits a small scale correction. Hypotheses are scored by inlier count, with mean residual as a tiebreak. Measured median residual is $22.8 \mu\text{m}$.

6.2 Lattice quantization and vault construction

Registered coordinates are quantized to a cubic lattice of pitch $\lambda = 4\bar{r} \approx 91 \mu\text{m}$, giving $N_c \approx 1.64 \times 10^5$ occupiable cells. Enrolment fuses $R = 7$ scans, keeping inclusions observed in

at least $\lceil 0.55R \rceil$ of them, which yields a persistent template $\mathcal{G}$ of $|\mathcal{G}| = 14.1 \pm 3.9$ points.

Because $\mathcal{G}$ is an unordered set subject to insertions and deletions rather than a bit string subject to substitutions, the Hamming-metric machinery of Proposition 1 does not apply, and we use a fuzzy vault [6]. The key is encoded in a degree- D polynomial Φ over $\text{GF}(q)$; each genuine lattice cell g_i contributes $(g_i, \Phi(g_i))$; C chaff points (c_j, y_j) with $y_j \neq \Phi(c_j)$ are added at unoccupied cells. A verifier whose probe recovers $g \geq D + 1$ genuine points can interpolate Φ by Reed–Solomon decoding.

6.3 Decoding condition and security

Chaff density creates false correspondences. A probe presenting P points collides with chaff at rate $\approx C/N_c$ per point, contributing s erroneous symbols. Reed–Solomon decoding of $g + s$ received symbols with dimension $D + 1$ succeeds when $s \leq (g + s - D - 1)/2$, i.e.

$$g \geq D + 1 + s. \quad (5)$$

Taking $g \sim \text{Bin}(|\mathcal{G}|, r)$ with recovery rate r and $s \sim \text{Bin}(P, C/N_c)$ gives the unlock probability we report. Brute force requires isolating a genuine $(D+1)$ -subset among $|\mathcal{G}| + C$ points, so

$$\text{sec} = \log_2 \binom{|\mathcal{G}| + C}{D + 1} - \log_2 \binom{|\mathcal{G}|}{D + 1}. \quad (6)$$

Equation (5) and Equation (6) pull against each other through both C and D , which is the tension Figure 7 visualizes.

7 Binding Protocol

7.1 The stone as a signer

The design move that closes G_1 and G_3 is to give the stone a keypair. Enrolment derives $\text{sk}_S = \text{KDF}(R \parallel \text{salt})$ from the vault key R and publishes pk_S . No party stores sk_S : it exists only inside an attested scanner, for the duration of a verification, and only when the physical stone is present. Every provenance operation carries a signature under sk_S , so the record cannot move without the object.

7.2 Enrolment

A certifying laboratory acquires R scans across all three modalities, constructs the Stage-A code and the Stage-B vault, derives sk_S , and publishes

$$\text{rec}_S = (\text{pk}_S, H(\mathcal{V}), \text{cls}, H(\text{report}), \tau, \sigma_{\text{lab}}, \sigma_S),$$

where $\mathcal{V}$ is the vault (stored off-chain, content-addressed), cls is the M_2 material-class attestation, and τ is a monotonic counter. The record is co-signed by the laboratory and by the stone. Enrolment is the trust anchor and the point of maximum exposure to A_8 , which is why Section 8 requires k -of- n independent laboratories for high-value stones.

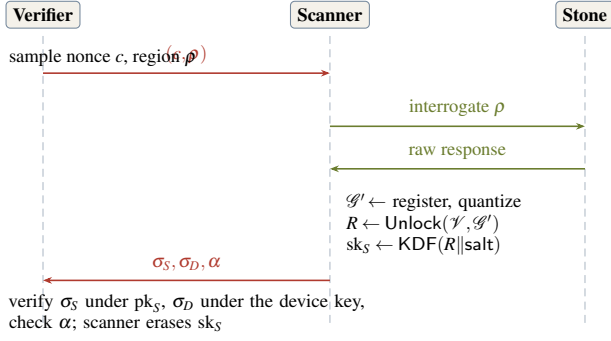


Figure 2: Verification. The verifier chooses both a nonce and an interrogation region, so a stored measurement cannot answer, and the secret key exists only transiently inside the attested device.

7.3 Challenged interrogation

Freshness (G_2) and resistance to a compromised sensor (A_5) come from making the measurement itself unpredictable. The verifier samples a region ρ : a sub-volume for M_3 confocal interrogation, together with an excitation wavelength and, where applicable, a polarization setting. The response is a hash over the measurement restricted to ρ , signed under both sk_S and the device attestation key. Because the challenge space is large and the response depends on defect structure the protocol never publishes, a replayed measurement from a different stone fails, and a compromised scanner must synthesize a response for a region it has not seen.

The honest limit: an adversary with unrestricted physical access to the genuine stone can characterize it exhaustively and then answer any challenge from a compromised device. Challenged interrogation defends against remote and partial-knowledge adversaries, not against $\mathcal{A}_{\text{ins}}$ holding the stone. We regard that residual as irreducible.

7.4 Transfer, cloning, and re-cutting

Transfer (G_3). A custody transfer is a transaction signed by the sender, the receiver, and the stone, incrementing τ . Absent the physical stone, no valid transfer exists.

Clone detection (G_5). Cryptography cannot prevent physical cloning; it can make cloning visible. Each verification appends a signed $(\tau, \text{time}, \text{location}, \text{device})$ tuple. Two stones sharing pk_S necessarily produce either a counter fork or a spatiotemporally impossible pair of verifications. This is physical double-spend detection, with the same guarantee structure: detection and attribution after the fact rather than prevention.

Re-cutting (G_6). Section 9 shows re-polishing drops vault unlock probability from 0.97 to 0.09, so treating re-cut stones as transparently re-verifiable is untenable. We instead require an explicit transition: while the stone still satisfies the old key, it signs a transition message committing to the new enrolment, witnessed by a laboratory. A re-cut performed

without this ceremony produces a stone with no valid record, which is the correct outcome, since an unrecorded re-cut is indistinguishable from A_7 .

8 Security Analysis

A_1 Record forgery.

Requires a laboratory signature. Reduces to laboratory key compromise; mitigated by k -of- n certification and by public transparency logs over enrolments.

A_2 Substitution.

The substitute must unlock the vault. Section 9 measures impostor unlock at 9.5×10^{-5} for the chosen parameters, and the M_2 gate independently rejects grown material at 99.8% accuracy. The 2024 incident [28] fails at both layers.

A_3 Cloning.

Physical cloning of M_1 requires placing ≥ 20 inclusions at specified positions inside a grown crystal; of M_3 , atomic-scale placement of $\sim 10^4$ colour centres. Neither is prevented by the protocol; both are detected by G_5 .

A_4 Replay.

Defeated by nonce and region binding, provided the challenge space is large relative to the adversary’s stored measurements.

A_5 Oracle compromise.

Constrained but not eliminated. Device attestation plus unpredictable regions forces the adversary to synthesize physically consistent responses. We consider this the weakest link and treat k -of- n multi-device verification as mandatory above a value threshold.

A_6 Template inversion.

The gravest finding of Section 9. Vault security Equation (6) assumes the adversary cannot distinguish chaff from genuine points. An adversary who independently learns a large fraction of the inclusion map bypasses the vault entirely: at 90% knowledge, unlock probability is 0.49. Clarity plots published on grading reports are precisely such a partial map. Any deployment must therefore either withhold plot detail below the lattice pitch, or draw the vault’s genuine points from M_3 rather than M_1 . We regard this as a hard constraint on how the protocol may be deployed alongside existing disclosure practice.

A_7 Re-cut laundering.

Addressed by the transition ceremony; an unwitnessed re-cut invalidates the record.

A_8 Enrolment poisoning.

Mitigated by multi-laboratory enrolment and by requiring that enrolment scans be individually attested.

9 Evaluation

9.1 Methodology and its status

We simulate 1,000 stones: 500 for fitting projections and quantization thresholds, 500 held out. Each stone has $K \sim$

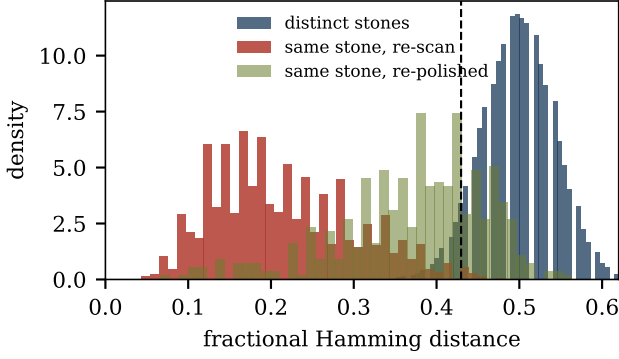


Figure 3: Stage-A Hamming distance distributions ($n=135$). Genuine and impostor populations separate cleanly, but re-polished stones sit between them, foreshadowing the lifecycle problem addressed in Section 7.

Poisson(18) inclusions, 60% scattered through an anisotropic ellipsoid and 40% concentrated on one or two growth planes, which imposes the spatial correlation that suppresses true entropy relative to Equation (1). Each acquisition applies a random rigid pose, per-axis localization noise $\sigma = 8\mu\text{m}$, size-dependent detection failure (base rate 6%), Poisson(0.6) spurious detections, and multiplicative size noise. Enrolment uses $R = 7$ scans and probing uses 5. Re-polishing removes inclusions within the outer 12% of the radius and leaves the survivors rigid with respect to one another, since polishing removes surface material rather than deforming the lattice. Grown stones carry an $[\text{SiV}]^-$ and 596/597 nm signature in the PL block.

These are simulation results on synthetic data. They are predictions about what a physical study would find, conditional on a generative model that we chose. They are not measurements of diamonds, and no claim here should be cited as such.

9.2 Stage A: screening

Figure 3 shows the Hamming geometry. Impostor distance concentrates at 0.498 ± 0.044 , as a well-balanced code should. Genuine re-scan distance is 0.172 at $n = 64$, rising to 0.216 as lower-reliability bits are admitted.

The supervised step matters. Within-class whitening lowers the equal error rate from 8.9×10^{-2} to 1.9×10^{-2} on an identical raw descriptor (Figure 4), which isolates the gain to the projection rather than to feature engineering. At a screening radius $t/n = 0.43$ with false rejection 10^{-2} , false acceptance is 6.9×10^{-2} , adequate for shortlisting a registry and inadequate for anything else.

Finding 1: the alignment-free descriptor cannot bind. Applying Proposition 1 to the measured operating points, the best extractable key length over all n is -124 bits: the correlation-corrected entropy (54.6 bits at $n = 64$, 106.4 at $n = 135$) never exceeds the sum of error-correction leakage

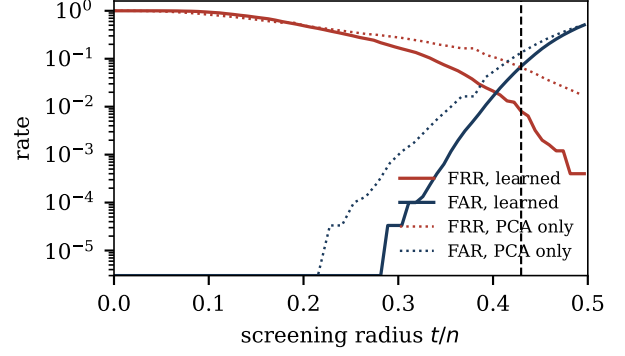


Figure 4: Screening error rates. Learned within-class whitening improves the equal error rate by $4.7\times$ over unsupervised whitening on the identical descriptor.

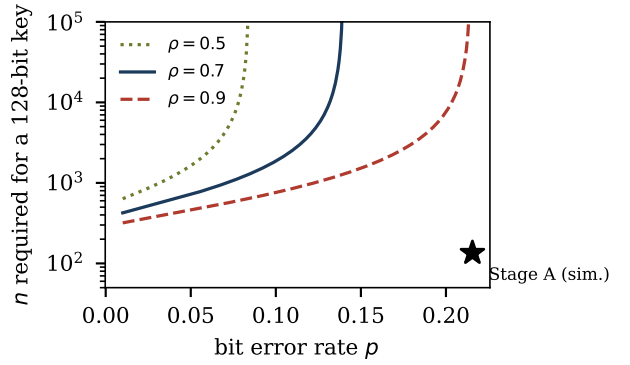


Figure 5: Feasible region from Proposition 1 at $\ell=128$. The measured Stage-A operating point (star) lies below every contour, so no code length makes a single-stage fuzzy extractor over these invariants viable.

and extractor loss. Figure 5 places the operating point far below every feasible contour. This is a structural result rather than a tuning failure. Global invariants of a sparse point set discard the very information that distinguishes stones, and no amount of downstream cryptography recovers it.

Finding 2: the material-class gate must be spectroscopic.

All 200 simulated CVD substitutes pass the geometric screen, with nearest-template distance 0.365 ± 0.018 that is indistinguishable from any other impostor. This is correct behaviour rather than a defect: a grown stone’s inclusion constellation is a different constellation, carrying no signature of how it was made. The PL sub-block alone separates the classes at 99.8% accuracy. Systems that infer growth origin from geometry or from proportions are inferring it from a variable that does not carry the information.

9.3 Stage B: key extraction

Registration succeeds with median residual $22.8\mu\text{m}$ across 195 evaluated stones, yielding persistent templates of $|\mathcal{S}| = 14.1 \pm 3.9$ points and a $91\mu\text{m}$ lattice with 1.64×10^5 cells.

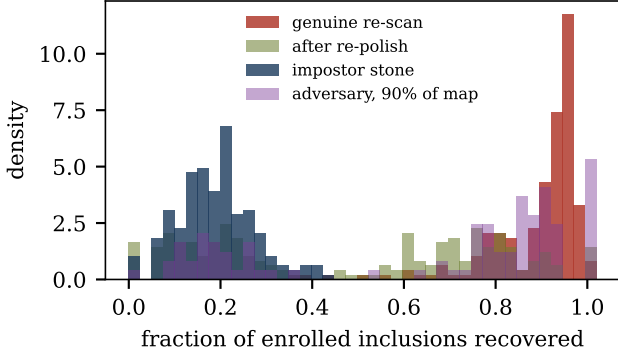


Figure 6: Fraction of enrolled inclusions recovered. Genuine re-scans recover 0.892; impostors recover 0.187, driven by lattice collisions rather than by true correspondence; re-polished stones recover 0.479; an adversary holding 90% of the map recovers 0.663.

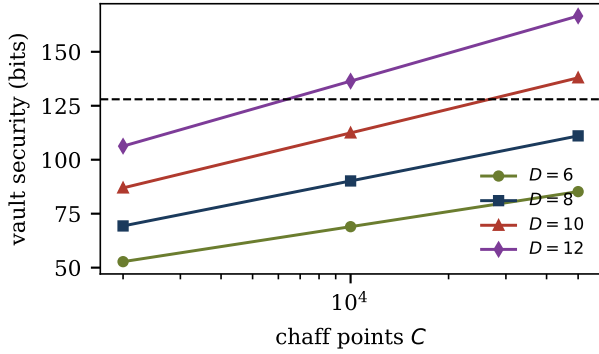


Figure 7: Vault security Equation (6) against chaff count. Higher D and higher C both raise security and both lower the unlock probability through Equation (5), so the usable region is bounded above by Figure 6.

Table 4 gives the joint operating characteristic. The best point satisfying $\Pr[\text{unlock}] \geq 0.95$ is $C = 10^4$, $D = 8$, delivering 90.2-bit security with 0.970 unlock and false unlock 9.5×10^{-5} . Reaching 128 bits at this $|\mathcal{G}|$ costs unlock probability: $C = 10^4$, $D = 12$ gives 136.4 bits at 0.288 unlock, which is unusable.

Finding 3: security is an instrument specification. Figure 8 sweeps $|\mathcal{G}|$ holding the recovery rate fixed. The achievable security at 95% unlock rises from 66.8 bits at $|\mathcal{G}| = 10$ to 116.5 at 18 and 141.7 at 22, crossing 128 bits at $|\mathcal{G}| \approx 20$. The binding question is therefore not which cryptographic construction to use but how many inclusions a mapper can resolve *reproducibly*, which is a specification a laboratory can procure against. We note that commercial systems already claim micron-level detection on rough [33], so the constraint plausibly lies in reproducibility across instruments and settings rather than in raw sensitivity.

C	D	sec. (bits)	$\Pr[\text{unlock}]$	$\Pr[\text{false}]$
2,000	8	69.3	0.995	1.9×10^{-4}
10,000	8	90.2	0.970	9.5×10^{-5}
10,000	10	112.4	0.779	8.2×10^{-7}
10,000	12	136.4	0.288	1.5×10^{-9}
50,000	10	138.0	0.106	1.2×10^{-8}

Table 4: Vault operating points at $|\mathcal{G}| = 14$. Security and usability are in direct conflict at this template size.

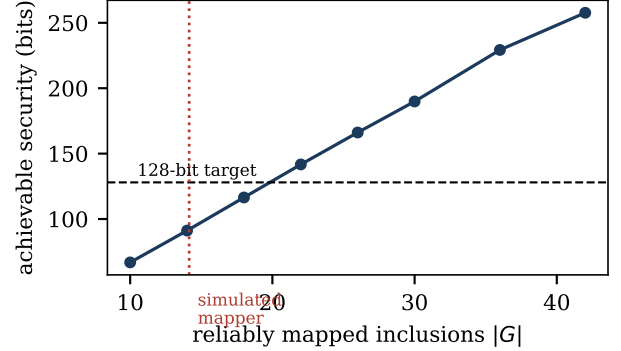


Figure 8: Achievable security against template size at fixed recovery rate, maximized over (C, D) subject to $\Pr[\text{unlock}] \geq 0.95$. Twenty reliably mapped inclusions is the 128-bit threshold.

9.4 Adversarial stress tests

Table 5 reports the partial-knowledge attack, in which an adversary learns a fraction of the true inclusion map and fabricates the remainder. Unlock probability is negligible below 50% knowledge, rises steeply through 75%, and reaches 0.487 at 90%. The response is not steep enough to be comfortable: the vault degrades gracefully from the adversary’s point of view, which is the wrong direction. This is the quantitative basis for the disclosure constraint in Section 8 and, in our view, the most important practical result in the paper.

9.5 Summary

Stage A screens and classifies but cannot bind. Stage B binds at 90 bits with current template sizes and at 128 bits given ≥ 20 reliably mapped inclusions. Neither stage survives re-polishing, which the protocol handles by ceremony rather than by robustness. And the confidentiality goal G_4 is load bearing in a way that the initial design underestimated.

10 Deployment Considerations

Throughput and cost. M_1 mapping is already in the manufacturing path, so its marginal cost is near zero for stones that pass through planning. M_2 adds a PL measurement at liquid-nitrogen temperature, routine in laboratories and impractical in retail. M_3 requires confocal microscopy and is

Map known	Recovery	Pr[unlock]
25%	0.235 ± 0.163	5.9×10^{-4}
50%	0.266 ± 0.191	1.6×10^{-3}
75%	0.463 ± 0.307	7.7×10^{-2}
90%	0.663 ± 0.325	4.9×10^{-1}

Table 5: Partial-knowledge attack at $C = 10^4$, $D = 8$.

a certification-time and dispute-time instrument. This tiering keeps the common path cheap and reserves cost for the moments that carry value.

Interoperability with Tracr. ADAMAS is complementary rather than competing. Tracr establishes origin at source and follows the stone through manufacture; ADAMAS supplies the missing cryptographic binding at each checkpoint. Concretely, rec_S can be referenced from a Tracr asset record, and Tracr’s existing scanning integrations [31] are where the attested-device requirement would be introduced.

Privacy and trade secrecy. Dealers have legitimate reasons to hide inventory and margins. Verification therefore proves a predicate, that this stone corresponds to a record certified by an accredited laboratory and currently held by the asserting party, without revealing the record’s contents, using standard succinct proofs over the on-chain state [15, 14]. We deliberately keep the fuzzy-extraction computation outside the proof system, since proving a decoder in zero knowledge buys little here and costs a great deal.

11 Limitations and Threats to Validity

The generative model is the main risk. Real inclusion constellations are not our mixture of ellipsoid and growth planes. If real spatial correlation is stronger, extractable entropy falls and $|\mathcal{G}|$ must rise further; if inclusion counts in commercial goods are systematically lower than 18, the 128-bit threshold may be out of reach for many stones, which would push the design onto M_3 sooner. Our detection-failure and spurious-detection rates are plausible but unmeasured.

We do not simulate M_3 . The claim that NV sub-volumes supply $> 10^4$ bits rests on defect-density arithmetic rather than on a simulated readout pipeline with realistic confocal noise, drift, and charge-state instability. The root of trust is therefore the least validated layer.

Fuzzy vault weaknesses. Correlation attacks recover genuine points from multiple enrolments of the same secret [9]. Our protocol re-enrolls after re-cutting, which is exactly the situation such attacks exploit. Chaff must be regenerated under a fresh secret, and the number of lifetime enrolments must be bounded.

The linear encoder understates achievable performance. We used PCA followed by within-class whitening in place of a trained equivariant encoder to keep the ablation clean. A properly trained encoder should reduce BER and raise ρ , though Proposition 1 bounds how much that can help: the entropy of the underlying point set is what it is.

No physical validation. Nothing here has touched a diamond.

12 Research Plan

The results above convert cleanly into an experimental program with a gemological partner.

Phase 1 (months 1–4): measurement campaign. Acquire repeated inclusion maps of $\geq 1,000$ graded stones spanning clarity grades VVS through I, with ≥ 5 independent acquisitions per stone across ≥ 2 instruments to separate within-instrument from between-instrument variance. The primary deliverable is the empirical distribution of registration residual and of reproducibly mapped inclusion count, which Figure 8 shows is the quantity that determines everything else.

Phase 2 (months 3–8): matched-pair studies. Two subsets carry disproportionate value. First, before-and-after pairs across re-polishing, which determine whether G_6 can be relaxed from ceremony to robustness. Second, adversarially matched pairs, i.e. grown stones selected or produced to resemble specific natural targets in geometry, which measure the true $\mathcal{A}_{\text{fab}}$ false-accept rate rather than the random-impostor rate we report.

Phase 3 (months 6–12): learned encoder. Replace the linear projection with an $E(3)$ -equivariant graph encoder over the inclusion set, trained with an angular-margin identity loss, an explicit bit-decorrelation penalty targeting the ρ term of Proposition 1, and a reliability weighting derived from repeated acquisitions. Success criterion: BER below 5% at $\rho \geq 0.7$, which Table 3 shows suffices at $n \approx 720$.

Phase 4 (months 9–15): NV feasibility. Establish on a small sample whether an NV sub-volume can be re-located and re-measured reproducibly enough to support challenge-response, which is the single assumption the root of trust rests on and the one we have not tested.

Phase 5 (months 12–18): end-to-end pilot. Instrument a real custody chain with attested scanners and measure operational false-reject rates, which in biometric systems invariably exceed laboratory estimates.

13 Conclusion

Ledger-based provenance secures the record and leaves the correspondence between record and object to a trusted operator. We have argued that this gap is the substantive security problem in diamond provenance, that closing it fully is impossible, and that the reachable goal is a reduction of forgery to atomic-scale replication, instrument compromise, or possession of the genuine stone.

ADAMAS pursues that reduction by deriving a key from crystallographic defect structure and making possession of the stone a precondition for every provenance operation. Simulation on 1,000 synthetic stones supports the architecture and sharpens it in three ways we did not anticipate. Alignment-free invariants are structurally incapable of cryptographic binding, so registration is not an implementation detail but the core of the design. Growth origin is invisible to geometry and must be gated spectroscopically. And the threshold for 128-bit security is roughly twenty reproducibly mapped inclusions, which makes the security level a procurement specification rather than a protocol choice.

The finding we would most want tested first is the disclosure risk. If an adversary holding ninety percent of an inclusion map unlocks the vault half the time, then publishing clarity plots and deploying an inclusion-derived key are mutually incompatible, and the field should learn that before building the system rather than after.

Reproducibility

The simulation harness that produced every figure and table is a single self-contained Python script with a fixed seed, released alongside this manuscript.

References

- [1] R. Pappu, B. Recht, J. Taylor, and N. Gershenfeld. Physical one-way functions. *Science*, 297(5589):2026–2030, 2002.
- [2] B. Gassend, D. Clarke, M. van Dijk, and S. Devadas. Silicon physical random functions. In *ACM CCS*, 2002.
- [3] G. E. Suh and S. Devadas. Physical unclonable functions for device authentication and secret key generation. In *DAC*, 2007.
- [4] Y. Dodis, L. Reyzin, and A. Smith. Fuzzy extractors: How to generate strong keys from biometrics and other noisy data. *SIAM Journal on Computing*, 38(1):97–139, 2008.
- [5] A. Juels and M. Wattenberg. A fuzzy commitment scheme. In *ACM CCS*, 1999.
- [6] A. Juels and M. Sudan. A fuzzy vault scheme. *Designs, Codes and Cryptography*, 38(2):237–257, 2006.
- [7] R. Maes, P. Tuyls, and I. Verbauwhede. A soft decision helper data algorithm for SRAM PUFs. In *IEEE ISIT*, 2009.
- [8] U. Rührmair, F. Sehnke, J. Sölter, G. Dror, S. Devadas, and J. Schmidhuber. Modeling attacks on physical unclonable functions. In *ACM CCS*, 2010.
- [9] W. J. Scheirer and T. E. Boulton. Cracking fuzzy vaults and biometric encryption. In *Biometrics Symposium*, 2007.
- [10] M. S. Turan et al. Recommendation for the entropy sources used for random bit generation. NIST SP 800-90B, 2018.
- [11] S. Nakamoto. Bitcoin: A peer-to-peer electronic cash system. 2008.
- [12] V. Buterin. Ethereum: A next-generation smart contract and decentralized application platform. 2014.
- [13] F. Zhang, E. Cecchetti, K. Croman, A. Juels, and E. Shi. Town Crier: An authenticated data feed for smart contracts. In *ACM CCS*, 2016.
- [14] E. Ben-Sasson, A. Chiesa, C. Garman, M. Green, I. Miers, E. Tromer, and M. Virza. Zerocash: Decentralized anonymous payments from Bitcoin. In *IEEE S&P*, 2014.
- [15] J. Groth. On the size of pairing-based non-interactive arguments. In *EUROCRYPT*, 2016.
- [16] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. L. Hollenberg. The nitrogen-vacancy colour centre in diamond. *Physics Reports*, 528(1):1–45, 2013.
- [17] G. Balasubramanian et al. Nanoscale imaging magnetometry with diamond spins under ambient conditions. *Nature*, 455:648–651, 2008.
- [18] P. C. Maurer et al. Room-temperature quantum bit memory exceeding one second. *Science*, 336(6086):1283–1286, 2012.
- [19] A. M. Zaitsev. *Optical Properties of Diamond: A Data Handbook*. Springer, 2001.
- [20] C. M. Breeding and J. E. Shigley. The “type” classification system of diamonds and its importance in gemology. *Gems & Gemology*, 45(2):96–111, 2009.
- [21] C. M. Breeding and W. Wang. Occurrence of the Si-V defect center in natural colorless gem diamonds. *Diamond and Related Materials*, 17(7–10):1335–1344, 2008.
- [22] P. M. Martineau et al. Identification of synthetic diamond grown using chemical vapor deposition (CVD). *Gems & Gemology*, 40(1):2–25, 2004.
- [23] S. Eaton-Magaña and C. M. Breeding. An introduction to photoluminescence spectroscopy for diamond and its applications in gemology. *Gems & Gemology*, 52(1), 2016.
- [24] S. Eaton-Magaña et al. Shining a light on gemstone properties: An exploration of photoluminescence spectroscopy. *Gems & Gemology*, 60(4), 2024.
- [25] S. Eaton-Magaña and J. E. Shigley. Observations on CVD-grown synthetic diamonds: A review. *Gems & Gemology*, 52(3), 2016.
- [26] GIA Laboratory. CVD-grown diamond with few diagnostic features. *Gems & Gemology Lab Notes*, Fall 2023.
- [27] GIA Laboratory. Silicon in natural type IIa and type IaB diamond. *Gems & Gemology Lab Notes*, Winter 2013.
- [28] International Gemological Institute. IGI uncovers fraudulent 6 ct laboratory-grown diamond bearing a natural diamond inscription. 2024. <https://www.igi.org/igi-uncovers-fraudulent-6ct-lab-grown-diamond-with-natural-inscription/>

- [29] Gemological Institute of America. GIA acquires 30% shareholding in diamond provenance blockchain platform Tracr. Press release, May 29, 2026. <https://www.gia.edu/gia-news-press/gia-and-tracr>
- [30] De Beers Group. GIA acquires 30 percent shareholding in diamond provenance blockchain platform Tracr. May 29, 2026.
- [31] De Beers Group. Tracr: Diamond traceability platform. Case study, 2024.
- [32] The Next Web. De Beers weaponises blockchain to fight lab-grown diamonds. June 2026.
- [33] Sarine Technologies. Galaxy Ultra inclusion mapping system. Product documentation.
- [34] Sarine Technologies. Technology overview: Galaxy, DiaExpert, DiaMension.
- [35] Sarine Technologies. Sarine Diamond Journey: traceability and polished validation.
- [36] Authors of “High-dimensional anticounterfeiting nanodiamonds authenticated with deep metric learning.” *Nature Communications*, 15, 2024.
- [37] Authors of “Manufacturing unclonable anti-counterfeiting labels using robust diamond microparticles on heterogeneous substrates.” 2023.
- [38] V. G. Satorras, E. Hoogeboom, and M. Welling. E(n) equivariant graph neural networks. In *ICML*, 2021.
- [39] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, and A. Smola. Deep sets. In *NeurIPS*, 2017.
- [40] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *CVPR*, 2017.
- [41] J. Deng, J. Guo, N. Xue, and S. Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *CVPR*, 2019.

A Protocol Pseudocode

Algorithm 1 Enroll(S) — executed by a certifying laboratory

- 1: acquire M_1 scans $\{s_1, \dots, s_R\}$, M_2 spectrum π , M_3 reference map v
 - 2: $\text{cls} \leftarrow \text{ClassGate}(\pi)$ $\triangleright$ natural / CVD / HPHT / treated
 - 3: **if** $\text{cls} \neq \text{natural}$ and origin claim is natural **then abort**
 - 4: $\mathcal{G} \leftarrow \text{FuseTemplate}(s_1, \dots, s_R)$ $\triangleright$ register, quantize, keep persistent points
 - 5: **if** $|\mathcal{G}| < G_{\min}$ **then abort** $\triangleright G_{\min} = 20$ per Figure 8
 - 6: sample $\Phi \xleftarrow{\$} \text{GF}(q)[x]$ of degree D ; $R_{\text{key}} \leftarrow \text{Encode}(\Phi)$
 - 7: $\mathcal{V} \leftarrow \{(g, \Phi(g)) : g \in \mathcal{G}\} \cup \text{Chaff}(C)$
 - 8: $\text{sk}_S \leftarrow \text{KDF}(R_{\text{key}} \parallel \text{salt})$; $\text{pk}_S \leftarrow \text{Pub}(\text{sk}_S)$
 - 9: $\text{rec}_S \leftarrow (\text{pk}_S, H(\mathcal{V}), \text{cls}, H(v), H(\text{report}), \tau=0)$
 - 10: publish rec_S signed by the laboratory and by sk_S ; **erase** $\text{sk}_S, R_{\text{key}}, \Phi$
-

Algorithm 2 Verify(S, rec_S) — verifier V and attested device $\mathcal{D}$

- 1: V : sample nonce $c \xleftarrow{\$} \{0, 1\}^{128}$ and interrogation region $\rho \xleftarrow{\$} \mathcal{R}$; send (c, ρ)
 - 2: $\mathcal{D}$: acquire M_1 scan s' and M_3 response v' restricted to ρ
 - 3: $\mathcal{D}$: $\mathcal{G}' \leftarrow \text{Register}(s', \text{helper})$; $\Phi' \leftarrow \text{RSDecode}(\mathcal{V}, \mathcal{G}')$
 - 4: **if** decoding fails **then return** $\perp$
 - 5: $\mathcal{D}$: $\text{sk}_S \leftarrow \text{KDF}(\text{Encode}(\Phi') \parallel \text{salt})$
 - 6: $\mathcal{D}$: $\sigma_S \leftarrow \text{Sign}_{\text{sk}_S}(c \parallel \rho \parallel H(v') \parallel \tau)$, $\sigma_D \leftarrow \text{Sign}_{\text{sk}_{\mathcal{D}}}(\sigma_S \parallel \alpha)$
 - 7: $\mathcal{D}$: **erase** sk_S, Φ' ; send $(\sigma_S, \sigma_D, \alpha, H(v'))$
 - 8: V : **return** $\text{Vf}_{\text{pk}_S}(\sigma_S) \wedge \text{Vf}_{\mathcal{D}}(\sigma_D) \wedge \text{CheckAttest}(\alpha) \wedge \text{ConsistentNV}(H(v'), \rho)$
-

Algorithm 3 Transfer($S, P_{\text{from}}, P_{\text{to}}$) and Recut(S)

- 1: **Transfer:** run Verify; on success submit $(\text{rec}_S, \tau+1, P_{\text{to}})$ signed by $\text{sk}_S, P_{\text{from}}$, and P_{to}
 - 2: ledger rejects any transfer whose τ is not the successor of the recorded value
 - 3: **Recut:** *before* polishing, run Verify and emit $\sigma_{\text{trans}} \leftarrow \text{Sign}_{\text{sk}_S}(\text{"intent-to-recut"} \parallel \text{plan})$
 - 4: after polishing, run Enroll to obtain $\text{pk}_{S'}$
 - 5: publish $(\text{pk}_S \rightarrow \text{pk}_{S'})$ carrying σ_{trans} and a laboratory witness signature
 - 6: mark pk_S retired; any later verification under pk_S is a clone signal
-

B Simulation Parameters

Parameter	Symbol	Value	Role
Stone half-dimension	R	$2500\,\mu\text{m}$	sets the position space
Inclusion count	K	Poisson(18), clipped [6,45]	entropy source size
Growth-plane fraction	—	0.40	imposes spatial correlation
Ellipsoid anisotropy	—	(1.00, 0.84, 0.63)	stabilizes the canonical frame
Localization noise	σ	$8\,\mu\text{m}$ per axis	mapper precision
Detection failure (base)	p_{miss}	0.06, size-scaled	deletions
Spurious detections	—	Poisson(0.6)	insertions
Size measurement noise	—	LogNormal(0, 0.12)	weight perturbation
Enrolment / probe scans	R_e/R_p	7/5	template quality
Re-polish depth	—	outer 12% of radius removed	lifecycle test
Population	—	500 train / 500 test	fit / evaluation split
Raw descriptor dimension	d	326	Stage-A input
PCA subspace	d'	275	rank reduction
Bit allocation thresholds	—	$\text{SNR} > \{1, 2, 4\}$	$k_j \in \{0, 1, 2, 3\}$
Correspondence tolerance	—	$120\,\mu\text{m}$	Stage-B matching
Lattice pitch	λ	$4\bar{r} \approx 91\,\mu\text{m}$	quantization
Rotation hypotheses	—	24	proper signed permutations
Random seed	—	20260721	reproducibility